$\underline{\sigma} = ?$ $\qquad \underline{\sigma} = \dfrac{\partial \psi}{\partial \underline{\varepsilon}}$

* The energy function is given by:

$$\psi = \frac{1}{2} K (tr\,\underline{\varepsilon})^2 + (1-D)\psi^0, \text{ where } \psi^0 = \mu \|dev\,\underline{\varepsilon}\|^2$$

$$\therefore \quad \psi = \frac{1}{2} K (tr\,\underline{\varepsilon})^2 + \mu(1-D)\|dev\,\underline{\varepsilon}\|^2$$

$$\underline{\sigma} = \frac{1}{2} K 2 (tr\,\underline{\varepsilon})\cdot \underline{I} + \mu(1-D) 2 \|dev\,\underline{\varepsilon}\| \cdot \frac{dev\,\underline{\varepsilon}}{\|dev\,\underline{\varepsilon}\|}$$

$$\underline{\sigma} = K\, tr\,\underline{\varepsilon}\, \underline{I} + 2\mu(1-D)\, dev\,\underline{\varepsilon}$$

→ The same derivation, step by step:

$$\frac{\partial \psi}{\partial \underline{\varepsilon}} = \frac{1}{2} K \frac{\partial (tr\,\underline{\varepsilon}^2)}{\partial tr\,\underline{\varepsilon}} \frac{\partial tr\,\underline{\varepsilon}}{\partial \underline{\varepsilon}} + \mu(1-D) \frac{\partial \left(\|dev\,\underline{\varepsilon}\|^2\right)}{\partial \|dev\,\underline{\varepsilon}\|} \cdot \frac{\partial \|dev\,\underline{\varepsilon}\|}{dev\,\underline{\varepsilon}} \cdot \frac{\partial dev\,\underline{\varepsilon}}{\partial \underline{\varepsilon}}$$

$$= \frac{1}{2} K 2\, tr\,\underline{\varepsilon}\, \underline{I} + \mu(1-D)\cdot 2 \|dev\,\underline{\varepsilon}\| \cdot \frac{dev\,\underline{\varepsilon}}{\|dev\,\underline{\varepsilon}\|} \cdot \underline{\underline{\mathbb{P}}} \qquad \left| \; dev\,\underline{\varepsilon} \cdot \underline{\underline{\mathbb{P}}} = dev\,\underline{\varepsilon} \right.$$

$$\underline{\sigma} = K\, tr\,\underline{\varepsilon}\, \underline{I} + 2\mu(1-D)\, dev\,\underline{\varepsilon}$$

→ The tangent modulus:

$$\underline{\underline{C}} \Rightarrow \quad \Delta \underline{\sigma} = \underline{\underline{C}} : \Delta \underline{\varepsilon} \quad \rightarrow \underline{\sigma}(\underline{\varepsilon}, D(\alpha))$$

$$\Delta \underline{\sigma} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} : \Delta \underline{\varepsilon} + \frac{\partial \underline{\sigma}}{\partial D} \cdot \frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}} : \Delta \underline{\varepsilon}$$

$$\underline{\underline{C}} = \frac{\Delta \underline{\sigma}}{\Delta \underline{\varepsilon}} \quad \Rightarrow \quad \underline{\underline{C}} = \frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} + \frac{\partial \underline{\sigma}}{\partial D} \frac{\partial D}{\partial \alpha} \otimes \frac{\partial \alpha}{\partial \underline{\varepsilon}}$$

* We need:

$$\frac{\partial \underline{\sigma}}{\partial \underline{\varepsilon}} = K \underline{I} \otimes \underline{I} + 2\mu(1-D)\underline{\underline{\mathbb{P}}}$$

$$\frac{\partial \underline{\sigma}}{\partial D} = -2\mu\, dev\,\underline{\varepsilon}$$

$$\frac{\partial D}{\partial \alpha} = \frac{D_\infty}{D_{s0}} \left(e^{-\frac{\alpha}{D_{s0}}} \right), \qquad \frac{\partial \alpha}{\partial \underline{\varepsilon}} = ?$$

Starting with $\dot{\alpha} = |\dot{\psi}^0|$ $\qquad \psi^0 = \mu \|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\|^2$

$$\dot{\psi}^0 = \left(\mu \|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\|^2\right)^{\cdot}$$

$\alpha(\varepsilon) \Rightarrow \dot{\alpha} = \dfrac{\partial \alpha}{\partial \underline{\underline{\varepsilon}}} : \dot{\underline{\underline{\varepsilon}}}$ $\qquad = \mu\, 2 \|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\| \cdot \dfrac{\partial \|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\|}{\partial \underline{\underline{\varepsilon}}} : \dot{\underline{\underline{\varepsilon}}} = 2\mu \|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\| \dfrac{\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}}{\|\underline{\underline{dev}}\,\underline{\underline{\varepsilon}}\|} : \dot{\underline{\underline{\varepsilon}}}$

$$\dot{\alpha} = \frac{\partial \alpha}{\partial \underline{\underline{\varepsilon}}} : \dot{\underline{\underline{\varepsilon}}} = |\dot{\psi}^0| \qquad \Rightarrow \boxed{\dot{\psi}^0 = 2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} : \dot{\underline{\underline{\varepsilon}}}}$$

$$\frac{\partial \alpha}{\partial \underline{\underline{\varepsilon}}} : \dot{\underline{\underline{\varepsilon}}} = \left| 2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} : \dot{\underline{\underline{\varepsilon}}} \right|$$

$$\frac{\partial \alpha}{\partial \underline{\underline{\varepsilon}}} : \dot{\underline{\underline{\varepsilon}}} = \begin{cases} 2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} : \dot{\underline{\underline{\varepsilon}}} & , \dot{\psi}^0 \geq 0 \\ -2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} : \dot{\underline{\underline{\varepsilon}}} & , \dot{\psi}^0 < 0 \end{cases}$$

$$\boxed{\frac{\partial \alpha}{\partial \underline{\underline{\varepsilon}}} = \begin{cases} 2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} & , \dot{\psi}^0 \geq 0 \\ -2\mu\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} & , \dot{\psi}^0 < 0 \end{cases}}$$

$$\boxed{\begin{aligned} \underline{\underline{\mathbb{C}}} &= K\, \underline{\underline{I}} \otimes \underline{\underline{I}} + 2\mu(1-D)\underline{\underline{\mathbb{P}}} + \underline{\underline{\mathbb{C}}}^D \\ \underline{\underline{\mathbb{C}}}^D &= \begin{cases} -4\mu^2 \dfrac{\partial D}{\partial \alpha}\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} \otimes \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} & , \dot{\psi}^0 \geq 0 \\ 4\mu^2 \dfrac{\partial D}{\partial \alpha}\, \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} \otimes \underline{\underline{dev}}\,\underline{\underline{\varepsilon}} & , \dot{\psi}^0 < 0 \end{cases} \end{aligned}}$$

* Time integration of $\dot{\alpha} = |\dot{\psi}^0|$

$$\int_{t_n}^{t_{n+1}} \dot{\alpha}\, dt = \int_{t_n}^{t_{n+1}} |\dot{\psi}^0|\, dt$$

$$\frac{\alpha_{n+1} - \alpha_n}{\Delta t} = \begin{cases} \displaystyle\int_{t_n}^{t_{n+1}} \dot{\psi}^0 dt & , \int_{t_n}^{t_{n+1}} \dot{\psi}^0 \geq 0 \\ \displaystyle\int_{t_n}^{t_{n+1}} -\dot{\psi}^0 dt & , \int_{t_n}^{t_{n+1}} \dot{\psi}^0 < 0 \end{cases} = \begin{cases} \dfrac{\psi^0_{n+1} - \psi^0_n}{\Delta t} & , \dfrac{\dot{\psi}^0_{n+1} - \dot{\psi}^0_n}{\Delta t} \geq 0 \quad (>0) \\ \dfrac{-(\psi^0_{n+1} - \psi^0_n)}{\Delta t} & , \dfrac{\dot{\psi}^0_{n+1} - \dot{\psi}^0_n}{\Delta t} < 0 \quad (>0) \end{cases}$$

$$\boxed{\alpha_{n+1} = \alpha_n + |\psi^0_{n+1} - \psi^0_n|}$$